



Disturbance observer based multi-variable control of ball mill grinding circuits

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ABSTRACT

Ball mill grinding circuits are essentially multi-variable systems characterized with couplings, time-varying parameters and time delays. The control schemes in previous literatures, including detuned multi-loop PID control, model predictive control (MPC), robust control, adaptive control, and so on, demonstrate limited abilities in control ball mill grinding process in the presence of strong disturbances. The reason is that they do not handle the disturbances directly by controller design. To this end, a disturbance observer based multi-variable control (DOMC) scheme is developed to control a two-input-two-output ball mill grinding circuit. The systems considered here are with lumped disturbances which include external disturbances, such as the variations of ore hardness and feed particle size, and internal disturbances, such as model mismatches and coupling effects. The proposed control scheme consists of two compound controllers, one for the loop of product particle size and the other for the loop of circulating load. Each controller includes a PI feedback part and a feed-forward compensation part for the disturbances by using a disturbance observer (DOB). A rigorous analysis is also given to show the reason why the DOB can effectively suppress the disturbances. Performance of the proposed scheme is compared with those of the MPC and multi-loop PI schemes in the cases of model mismatches and strong external disturbances, respectively. The simulation results demonstrate that the proposed method has a better disturbance rejection property than those of the MPC and PI methods in controlling ball mill grinding circuits.

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1. Introduction

Ball mill grinding circuit is one of the most important mineral processing units in concentrator plants. The product particle size of the grinding circuit has a great influence on the recovery rate of the valuable minerals and the final tailing discharge. Accordingly, an appropriate control of the grinding circuits is especially significant to reduce the energy losses and enhance the efficiency of the subsequent processing. However, essentially, ball mill grinding circuit is a typical multi-input-multi-output (MIMO) system characterized with high couplings, time-varying parameters, strong external disturbances and time delays, which cause undesirable influences on process behaviors.

Due to the complexities and difficulties, the controller design of the grinding circuit has attracted much attention of many industrial practitioners. Many research papers [1–10] have been published on the development of control strategies for grinding circuits. Traditionally, the grinding circuits are controlled by multi-loop PID controllers [1]. The detuned controllers has the advantage of easy parameter tuning. Another popular approach is model predictive control (MPC) [1,2,6,10]. MPC scheme possesses good capabilities of decoupling and dealing with time delays. Also many other advanced control schemes, such as robust control [3], adap-

tive control [4], neuro-control [5], expert control [7,8] and supervisory control [8,9], have been proposed in control literatures, and they improve the performances of grinding process from different aspects.

In grinding process, the disturbances have a great influence on the control performance of the closed-loop system. The product particle size and circulating load are sensitive to various disturbances which can be divided into external and internal ones. The external disturbances include the variations of ore hardness and feed particle size, and so on, while the internal disturbances are generally caused by model mismatches and coupling effects. Variations of the ore hardness and feed particle size may cause continuous fluctuations of the product particle size. Model mismatches and coupling effects may affect the dynamic features and even result in unstable control of the grinding circuits.

It should be pointed out that the above mentioned schemes demonstrate limited abilities in control ball mill grinding process in the presence of strong disturbances. The reason is that they do not deal with disturbances directly by controller design. Usually these controllers can not react directly and fast to reject these disturbances, although these controllers can finally suppress the disturbances through feedback regulation in a relatively slow way. Thus the controllers generally become sluggish when meeting strong disturbances.

To improve the system performance in such cases, a nature consideration is to introduce a feed-forward compensation part for the

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disturbances into the controller besides the conventional feedback part. Obviously, the most simple way is to obtain the model of the disturbance channel by mechanism analysis and cancel the disturbances directly in the control design. However, usually, it is impossible to obtain the precise mechanism model of disturbance in real grinding circuit applications. An effective solution is to develop disturbance estimation techniques. Disturbance observer (DOB) technique was originally presented by Ohnishi in 1987 [11]. Disturbance observer based control schemes for linear and nonlinear systems have been put forward and applied successively in many other control fields, e.g., robotic systems [12–14], hard drive systems [15–17], position system [18], missile system [19], general systems [20–23], etc. DOB has fine abilities in dealing with model mismatches and strong external disturbances. It does not rely on precise disturbance models. DOB is usually a two-input-one-output system which can generally be used for the feed-forward compensation design. The inputs are usually the manipulated variable and the controlled variable, while the output is an estimate of the disturbances. DOB estimates the lumped disturbance consisting of both the external and internal disturbances. The estimate of disturbance is employed in the feed-forward compensation design to attenuate the effects of the lumped disturbances.

Decoupling strategies in multi-variable control are always hot topics in the field of process control. As for MIMO systems, the variable pairing is generally based on relative gain array (RGA) analysis. Apart from paired manipulated variable, a certain controlled variable is also affected by other unpaired manipulated variables. From the point view of control, the effects caused by the unpaired manipulated variables can be viewed as a kind of internal disturbance. These coupling effects can be weakened by feed-forward compensation design based on DOB. The novelty of this paper lies in that a new strategy called disturbance observer based multi-variable control (DOMC) can not only overcome model mismatches as well as strong external disturbances, but also be used to attenuate the coupling effects of the ball mill grinding circuit (MIMO system).

In this paper, we try to develop a compound control strategy based on DOB technique for the ball mill grinding circuits in the presence of lumped disturbances. The control strategy consists of two compound controllers, one for the loop of product particle size and the other for the loop of circulating load. Each controller includes a PI feedback part and a feed-forward compensation part for the disturbances by using a disturbance observer (DOB). The standard DOB is not available since the systems here are with time delays. So we employ the modified DOBs with the considerations of the time delays [24]. Compared with other control strategies (such as detuned multi-loop PID control and inherently decoupling MPC algorithm), DOMC demonstrates the following remarkable features: (1) capability of handling model mismatches and strong external disturbances, (2) simplicity of parameter tuning, (3) low computation burden and complexities, and (4) easy industrial implementation, etc.

The rest of the paper is organized as follows. The process features of ball mill grinding circuits are illustrated in Section 2. In Section 3, conventional and modified disturbance observers are introduced at first, and then a multi-variable control scheme based on modified disturbance observer is described in details. Performance of the proposed method is compared with those of the MPC and PI methods in Section 4. Finally, conclusions are given in Section 5.

2. Process description

The grinding process is a size reduction operation used in the mineral industry to liberate the valuable minerals from the coarse ore. The typical plant flowsheet of a ball mill grinding circuit is

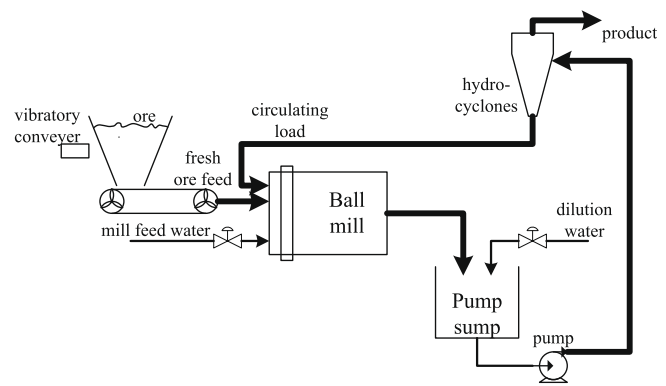


Fig. 1. Grinding circuit plant flowsheet.

shown in Fig. 1, which consists of a ball mill, a pump sump, hydro-cyclones and solids feeding conveyors.

When the ball mill works, the fresh coarse ore is fed into the mill together with water. The tumbling action of the balls within the revolving mill crushes the feed ore to finer sizes. The slurry containing the fine product is discharged from the mill to a pump sump, and after dilution they are pumped to hydro-cyclones for classification. The slurry is separated into two streams: an overflow stream containing the finer particles and an underflow stream containing the larger particles as the circulating load. The finer particles are the desired product, while the circulating load is recycled back to the ball mill for regrinding.

Product particle size is the most important controlled variable in a grinding circuit. Mine ore has to be ground to a specified size called the liberation size that the imbedded mineral particles are exposed for effective recovery in the downstream processing. It should be pointed out that grinding to a size much smaller than the liberation size may also bring about some drawbacks. Besides the higher power consumed in grinding, too fine particles or slimes are difficult to be recovered, which will also produce a larger volume of tailing discharge in the subsequent concentration and filtration process. The nominal values for the plant under study include: fresh ore feed rate 65 t/h, product particle size 70% – 200 mesh, circulating load 150 t/h and dilution water flow rate 45 m³/h.

Strong external disturbances in grinding circuits, such as variations of ore hardness and particle size, may cause continuous fluctuations of the product particle size. From the analysis of [25], we can obtain the results that ore hardness and the feed particle size affects the product particle size and circulating load through the channels with time delay and inertial element. The control objective of the systems can be presented as stable control of product particle size y_1 and circulating load y_2 . Both fresh ore feed rate u_1 and dilution water flow rate u_2 can be manipulated to regulate the two controlled variables [9]. The relationships among the controlled variables, manipulated variables and external disturbances in grinding process are illustrated clearly in Fig. 2.

3. Controller design

Detuned multi-loop PID control strategy is usually employed to handle the problem with control-loop interactions, with the advantages of easy tuning and commissioning. Dynamic decoupling usually requires exact process models and brings great difficulties to the industrial implementation, so static decoupling is used much more often. However, static decoupling degrades the control performance in real practice since it does not handle the dynamic coupling features of the process. Although MPC scheme possesses prominent decoupling features, it turns out some short-

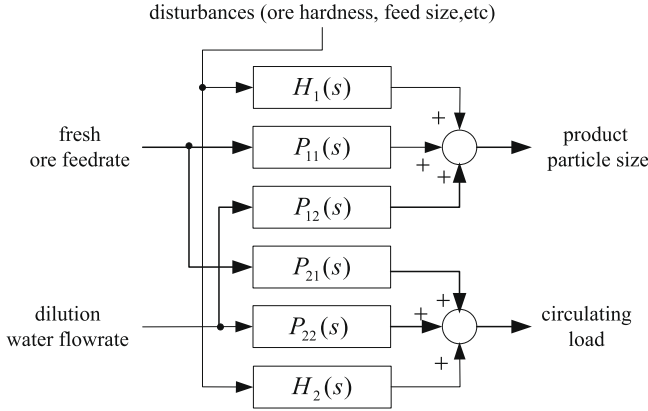


Fig. 2. The relationships among the variables in grinding process.

ages in the presences of strong external disturbances and severe model mismatches [10]. In this section, DOMC scheme is employed to control the coupling systems with model mismatches and strong external disturbances.

3.1. Conventional disturbance observer

The structure of disturbance observer (DOB) is generally based on the nominal model of the plant. The block diagram of the conventional disturbance observer [26] is presented in Fig. 3. In Fig. 3a, signals $C(s)$, $U(s)$, $D_{ex}(s)$, $N(s)$ and $Y(s)$ are the command of the controller, the manipulated variable, the external disturbance, the sensor noise and the controlled variable, respectively. Signals $\hat{D}(s)$ and $\hat{D}_f(s)$ are the estimates of the lumped disturbances before and after filtered by the low-pass filter $Q(s)$. $P(s)$ is the real plant to be controlled, while $G(s)$ is the nominal model of the plant. The transfer function of the output can be represented as follows

$$Y(s) = G_{cy}(s)C(s) + G_{dy}(s)D_{ex}(s) + G_{ny}(s)N(s), \quad (1)$$

where

$$G_{cy}(s) = \frac{P(s)G(s)}{G(s) + (P(s) - G(s))Q(s)}, \quad (2)$$

$$G_{dy}(s) = \frac{P(s)G(s)(1 - Q(s))}{G(s) + (P(s) - G(s))Q(s)}, \quad (3)$$

$$G_{ny}(s) = \frac{P(s)Q(s)}{G(s) + (P(s) - G(s))Q(s)}. \quad (4)$$

It can be seen from the above formulas that the design of DOB mainly depends on the design of $Q(s)$, and the time constants of the filter are the most meaningful parameters which determine the ability to resist disturbances.

Case 1: $Q(s) \rightarrow 1$.

If $Q(s) \rightarrow 1$, it can be concluded from (2)–(4) that

$$G_{cy}(s) \rightarrow G(s), G_{dy}(s) \rightarrow 0, G_{ny}(s) \rightarrow 1. \quad (5)$$

In this case, the dynamic features of the system are similar to that of $G(s)$.

Case 2: $Q(s) \rightarrow 0$.

If $Q(s) \rightarrow 0$, then

$$G_{cy}(s) \rightarrow P(s), G_{dy}(s) \rightarrow P(s), G_{ny}(s) \rightarrow 0. \quad (6)$$

In this case, the system is approximately open-loop without feed-back unit, thus the measurement noise can be filtered out.

Thus, usually, $Q(s)$ is designed to be a low-pass filter with steady-state gain of 1 so that:

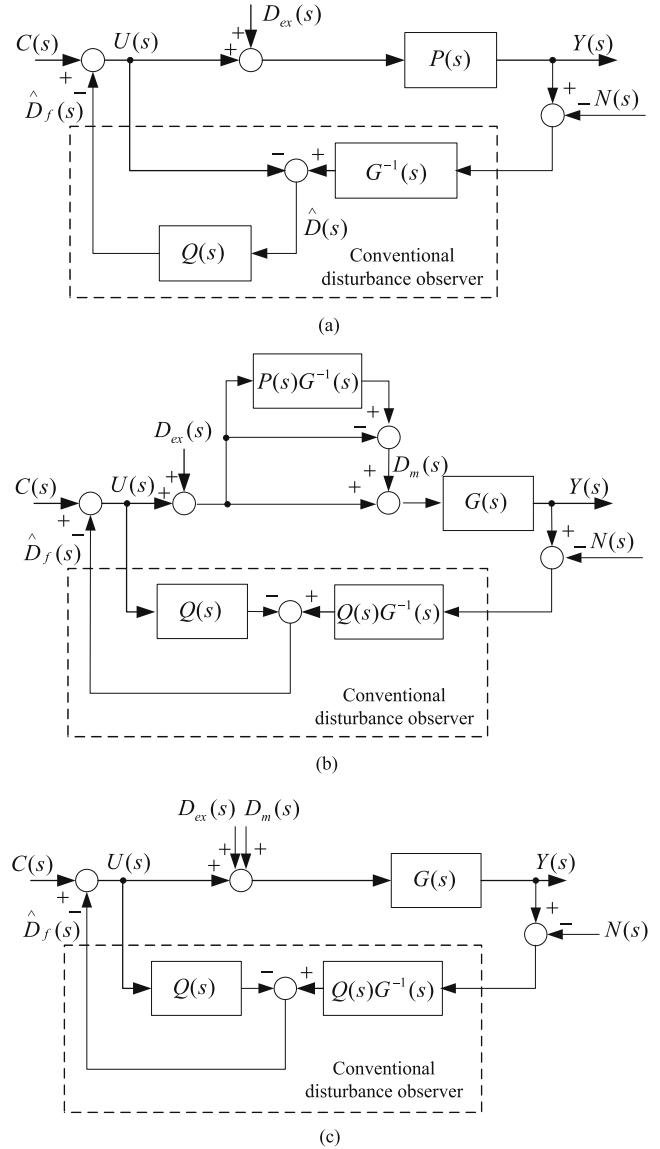


Fig. 3. Block diagram of the conventional disturbance observer.

- (1) in the domain of low-frequency, $Q(s)$ approaches to 1, guaranteeing that the estimate of the lumped disturbance approximately equals to the lumped disturbance. This means that the effects of the disturbance can be attenuated by a feed-forward compensation design based on DOB;
- (2) and in the domain of high-frequency, $Q(s)$ approximately equals to 0, which can ensure that the system possesses the features of the open-loop system, thus the high-frequency measurement noise can be filtered out.

However, the inverse of the nominal model $G^{-1}(s)$ is usually not proper and physically unrealizable. Considering the internal disturbance $D_m(s)$ caused by model mismatches, an equivalent block diagram of conventional disturbance observer is shown in Fig. 3b. To make sure that the control structure is realizable, $Q(s)G^{-1}(s)$ should be proper. In other words, the relative degree of $Q(s)$, i.e., the order difference between the denominator and the numerator, should be no less than that of the nominal model $G(s)$. The final equivalent block diagram of conventional disturbance observer is shown in Fig. 3c. Clearly, the lumped disturbance $D(s)$ consists of the external disturbance $D_{ex}(s)$ and internal disturbance $D_m(s)$ caused by model mismatches, i.e.,

$$D(s) = D_{ex}(s) + D_m(s), \quad (7)$$

where

$$D_m(s) = [P(s)G^{-1}(s) - 1](U(s) + D_{ex}(s)). \quad (8)$$

The internal disturbance $D_m(s)$ caused by model mismatches can also be estimated and compensated. The distinctive advantage of DOB lies in that it handles the disturbances without distinguishing between the external and internal disturbances.

3.2. Modified disturbance observer

In grinding process, the inverse function of time delay part is physical unrealizable, so the conventional disturbance observer is no longer available. To this end, here a structure of modified disturbance observer with the considerations of the time delays is employed to solve the above problems [24]. However, here the analysis of DOB is not similar to [24]. There are three differences between [24] and this paper. In [24], firstly the system is single-input-single-output, secondly the model mismatch is not considered, and thirdly the disturbances are added directly to the input of the plant. In this paper, the system is two-input-two-output, and we consider not only the internal disturbance caused by model mismatches, but also the other kind of internal disturbance caused by coupling effects. Moreover, the external disturbances are imposed on the output through disturbance channels.

Consider a two-input-two-output process which has the following form

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix} + \begin{bmatrix} D_{ex1}(s) \\ D_{ex2}(s) \end{bmatrix}. \quad (9)$$

with

$$P_{ij}(s) = p_{ij}(s)e^{-\tau_{ij}s} (i = 1, 2; j = 1, 2),$$

$$D_{exi}(s) = H_i(s)D_{ex}(s) (i = 1, 2).$$

where $U_i(s) (i = 1, 2)$ the manipulated variables, $Y_i(s) (i = 1, 2)$ the controlled variables, $D_{ex}(s)$ the external disturbance, $D_{exi}(s) (i = 1, 2)$ the effects of $D_{ex}(s)$ on $Y_i(s)$, $P_{ij}(s) (i = 1, 2; j = 1, 2)$ the models of the process channels, $p_{ij}(s) (i = 1, 2; j = 1, 2)$ the minimum-phase parts of $P_{ij}(s)$, and $H_i(s) (i = 1, 2)$ the disturbance models. Assume that u_1 is manipulated to control y_1 , also u_2 is manipulated to control y_2 . The structure of the modified disturbance observer for $u_1 - y_1$ loop is shown in Fig. 4. Note that the case of $u_2 - y_2$ loop is the similar which is omitted here. Clearly, the model $P_{11}(s)$ is separated into a minimum-phase part p_{11} and a time delay part $e^{-\tau_{11}s}$ as below

$$P_{11}(s) = p_{11}(s)e^{-\tau_{11}s}. \quad (10)$$

Similarly, the nominal model can be represented as

$$G_{11}(s) = g_{11}(s)e^{-\tau_{11}s}. \quad (11)$$

Compared with the conventional disturbance observer shown in Fig. 3c, we can find from Fig. 4a that the time delay part $e^{-\tau_{11}s}$ is inserted into the channel of the manipulated variable $U_1(s)$, and the minimum-phase part $g_{11}(s)$ is moved to the channel of the controlled variable $Y_1(s)$ to substitute $G(s)$.

Considering the effect of model mismatches, we obtain an equivalent block diagram shown in Fig. 4b. The final equivalent block diagram of the modified disturbance observer is shown in Fig. 4c. Clearly, the lumped disturbance $D_1(s)$ of $u_1 - y_1$ loop includes the external disturbance $D_{ex1}(s)$ and the internal disturbances consisting of $D_{m1}(s)$ caused by model mismatches and $D_{c1}(s)$ caused by coupling effects, i.e.,

$$D_1(s) = D_{ex1}(s) + D_{m1}(s) + D_{c1}(s), \quad (12)$$

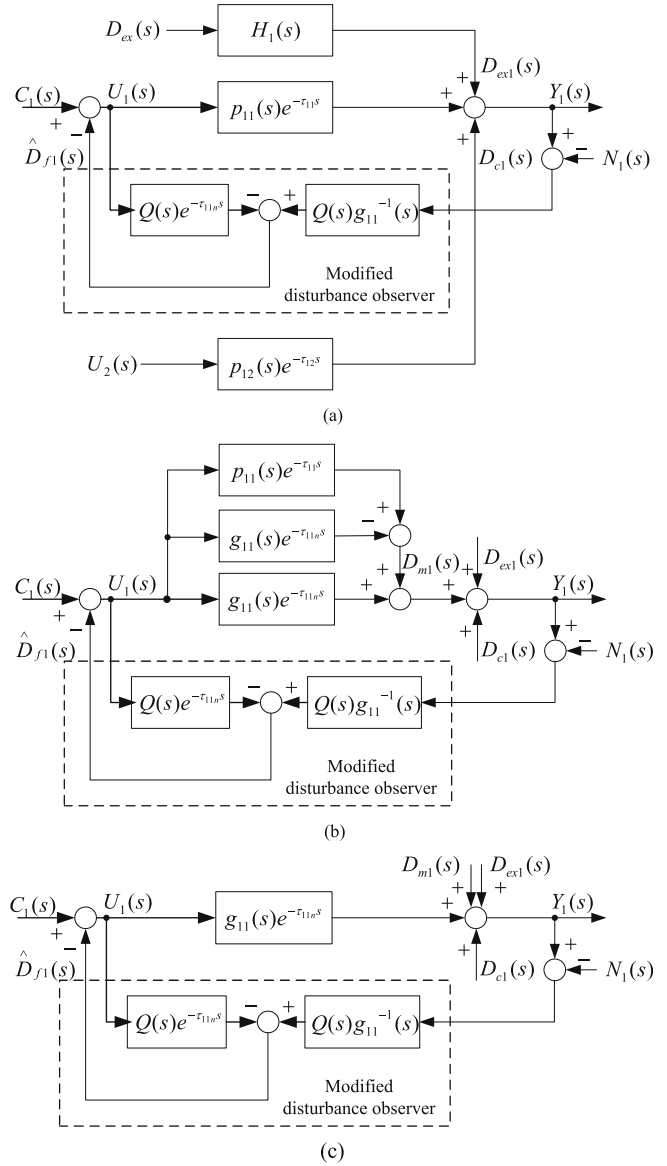


Fig. 4. Block diagram of the modified disturbance observer.

where

$$D_{m1}(s) = [p_{11}(s)e^{-\tau_{11}s} - g_{11}(s)e^{-\tau_{11}s}]U_1(s), \quad (13)$$

$$D_{c1}(s) = p_{12}(s)e^{-\tau_{12}s}U_2(s). \quad (14)$$

From Fig. 4c, one obtains

$$Y_1(s) = g_{11}(s)e^{-\tau_{11}s}U_1(s) + D_{m1}(s) + D_{ex1}(s) + D_{c1}(s). \quad (15)$$

Substituting (12) into (15), yields

$$Y_1(s) = g_{11}(s)e^{-\tau_{11}s}U_1(s) + D_1(s). \quad (16)$$

In the absence of sensor noise, i.e., $N_1(s) = 0$, it can be obtained from Fig. 4c

$$\hat{D}_{f1}(s) = Q(s)g_{11}^{-1}(s)Y_1(s) - Q(s)e^{-\tau_{11}s}U_1(s). \quad (17)$$

Substituting (16) into (17), yields

$$\hat{D}_{f1}(s) = Q(s)g_{11}^{-1}(s)D_1(s). \quad (18)$$

Define $\tilde{D}(s)$ as the error between the real value and the estimated value of the lumped disturbance, i.e.,

$$\tilde{D}_1(s) = D_1(s) - g_{11}(s)e^{-\tau_{11n}s}\tilde{D}_{f1}(s). \quad (19)$$

Substituting (18) into (19), yields

$$\tilde{D}_1(s) = [1 - Q(s)e^{-\tau_{11n}s}]D_1(s). \quad (20)$$

According to the Final-value Theorem, one obtains from (20)

$$\begin{aligned} \tilde{d}_1(\infty) &= \lim_{t \rightarrow \infty} \tilde{d}_1(t) = \lim_{s \rightarrow 0} s\tilde{D}_1(s) \\ &= \lim_{s \rightarrow 0} [1 - Q(s)e^{-\tau_{11n}s}] \lim_{s \rightarrow 0} sD_1(s) \\ &= \lim_{s \rightarrow 0} [1 - Q(s)e^{-\tau_{11n}s}] \lim_{t \rightarrow \infty} d_1(t) \\ &= \lim_{s \rightarrow 0} [1 - Q(s)e^{-\tau_{11n}s}] d_1(\infty). \end{aligned} \quad (21)$$

Obviously, if we select $Q(s)$ such that the steady-state gain of $Q(s)$ is 1, then one obtains from (21) that

$$\tilde{d}_1(\infty) = 0. \quad (22)$$

It can be found from (22) that the disturbances can be rejected. Hence, the modified disturbance observer is a practical approach to deal with the process with time delays.

3.3. Disturbance observer based multi-variable control for grinding circuit

Now we consider the application of DOB to the grinding circuit system. Note that the grinding circuit considered here is a multi-input-multi-output system, which means multiple DOBs should be designed. A new disturbance observer based multi-variable control scheme (DOMC) is proposed to solve couplings and overcome model mismatches as well as strong external disturbances in ball mill grinding circuits. The detailed design procedures for grinding process are described as follows.

Consider a typical two-input-two-output grinding circuit shown in Fig. 1. The manipulated variables are fresh ore feed rate u_1 (t/h) and dilution water flow rate u_2 (m³/h), while the controlled variables are product particle size y_1 (% – 200 mesh) and circulating load y_2 (t/h), respectively. According to the requirements of the real practice, the constraints of the manipulated variables and controlled variables are stated as follows: (1) the minimum and maximum values of the fresh ore feed rate are 60 t/h and

70 t/h, (2) the minimum and maximum values of the dilution water flow rate are 40 m³/h and 50 m³/h, (3) the minimum and maximum values of the product particle size are 68% and 72%, and (4) the minimum and maximum values of the circulating load are 140 t/h and 170 t/h.

Assume that the fresh ore feed rate u_1 is manipulated to control the product particle size y_1 , and the dilution water flow rate u_2 is manipulated to control the circulating load y_2 . In this case, the schematic diagram of DOMC is shown in Fig. 5. It can be observed that two PI controllers and two modified disturbance observers are employed here. PI controllers plus modified disturbance observers are selected to construct the DOMC structure. Here $Q(s)$ is selected as a first-order-low-pass filter with the steady-state gain of 1, which can be expressed as

$$Q(s) = \frac{1}{\lambda s + 1}. \quad (23)$$

We can tune the PI parameters and the time constants of the filter in DOB to achieve fine effects. According to the analysis in Section 3, some remarks on parameter tuning of the proposed method are listed as follows:

- As for the parameters tuning of PI feedback control part, we can refer to some general methods, such as Ziegler–Nichols, Tyreus–Luyben methods as well as method of trial and error according to experience.
- The time constants of the filters in DOBs are generally selected as a compromise value to satisfy the dynamics and stability of the closed-loop control systems. The general form and selection of binomial filters for $Q(s)$ have been suggested by Umeno and Hori [27].

4. Simulations and comparisons

For control study, plant test has been constructed to develop the input–output transfer function model as follows:

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix}, \quad (24)$$

where

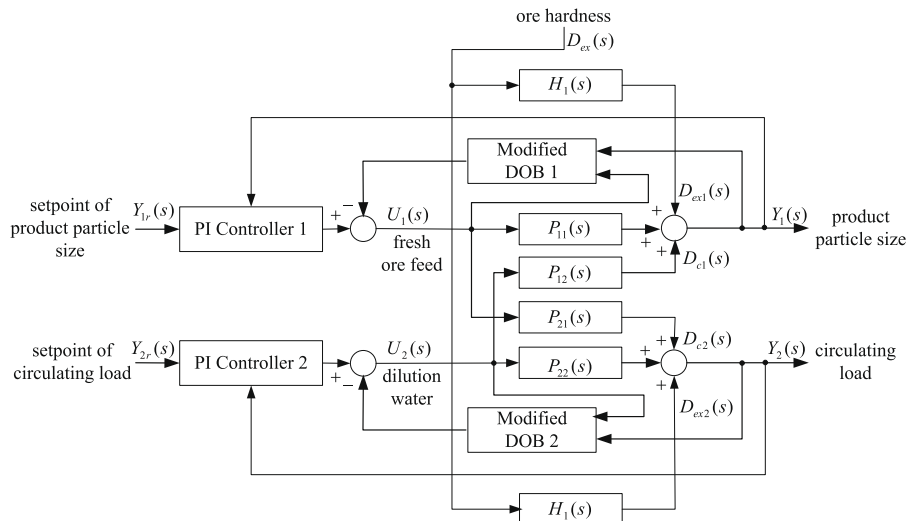


Fig. 5. The schematic diagram of DOMC applied in ball mill grinding circuit.

Table 1

Tuning parameters of the controllers for the grinding process.

Methods	Tuning parameters
DOMC	$K_{p1} = -1.4$, $T_{i1} = 3.2$ min, $K_{p2} = 0.35$, $T_{i2} = 4.2$ min, $\lambda_1 = 0.2$ min, $\lambda_2 = 1.7$ min
MPC	Sampling time $T_s = 0.5$ min, Prediction horizon (P) = 40, Control horizon (M) = 10, Weights on errors (Q) = $\text{diag}[1 \ 0.4]$, Weights on inputs (R) = $\text{diag}[0.5 \ 0.5]$
PI	$K_{p1} = -2$, $T_{i1} = 3.3$ min, $K_{p2} = 0.42$, $T_{i2} = 5.2$ min

$$G_{11}(s) = g_{11}(s)e^{-\tau_{11}s} = \frac{-0.58}{2.5s + 1}e^{-0.68s},$$

$$G_{12}(s) = g_{12}(s)e^{-\tau_{12}s} = \frac{4(1 - 0.9938e^{-0.47s})}{(2s + 1)(6s + 1)}e^{-0.2s},$$

$$G_{21}(s) = g_{21}(s)e^{-\tau_{21}s} = \frac{2.2}{6s + 1}e^{-0.6s},$$

$$G_{22}(s) = g_{22}(s)e^{-\tau_{22}s} = \frac{2.83}{3.5s + 1}e^{-0.13s}.$$

As for the process expressed in (24), a feasible variable pairing is $u_1 - y_1$ and $u_2 - y_2$ according to RGA analysis. The control performance of the process illustrated in (24) under the proposed method is compared with that under the other two widely used control

schemes: the MPC scheme and the multi-loop PI control scheme. DOMC and MPC controller are designed based on the model expressed in (24). In this section, simulation studies for set-point tracking in the case of model mismatches and external disturbance rejection are carried out, and the parameters of the three controllers are listed in Table 1.

4.1. Set-point tracking in the case of model mismatches

For simulation study, assume that the model of the operating plant holds the following form:

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} P_{11}(s) & P_{12}(s) \\ P_{21}(s) & P_{22}(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix}. \quad (25)$$

where

$$P_{11}(s) = \frac{-0.4e^{-0.5s}}{3.5s + 1},$$

$$P_{12}(s) = \frac{3.42e^{-0.13s}(1 - 0.994e^{-0.47s})}{(s + 1)(2s + 1)},$$

$$P_{21}(s) = \frac{0.8e^{-0.3s}}{3.2s + 1},$$

$$P_{22}(s) = \frac{2.5e^{-0.3s}}{5.5s + 1}.$$

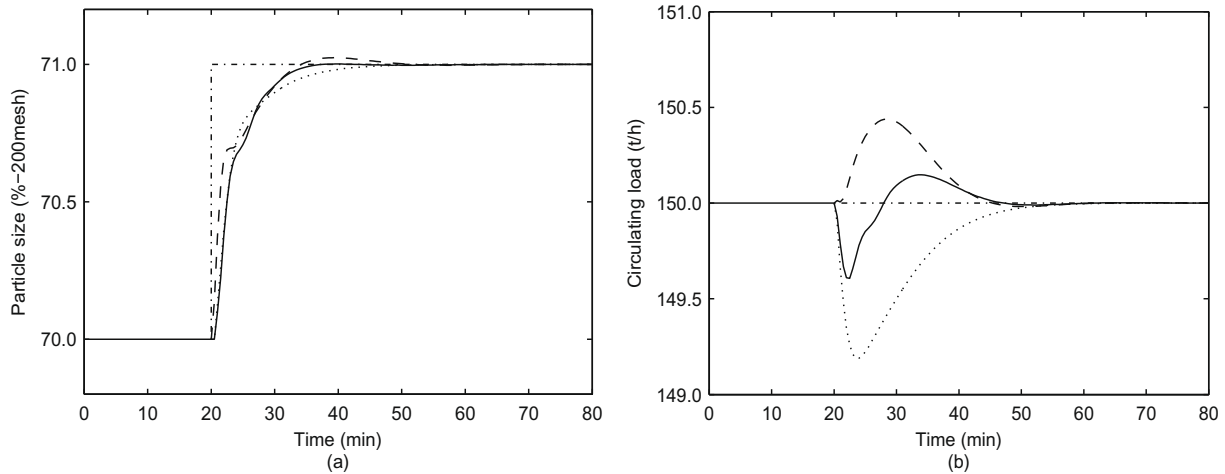


Fig. 6. Response curves of controlled variables for set-point change in particle size under DOMC (solid line), MPC (dash line) and PI (dot line) schemes in the presence of model mismatches: (a) particle size; (b) circulating load. The set-point change (dash-dot line) in particle size is from 70% to 71% – 200mesh.

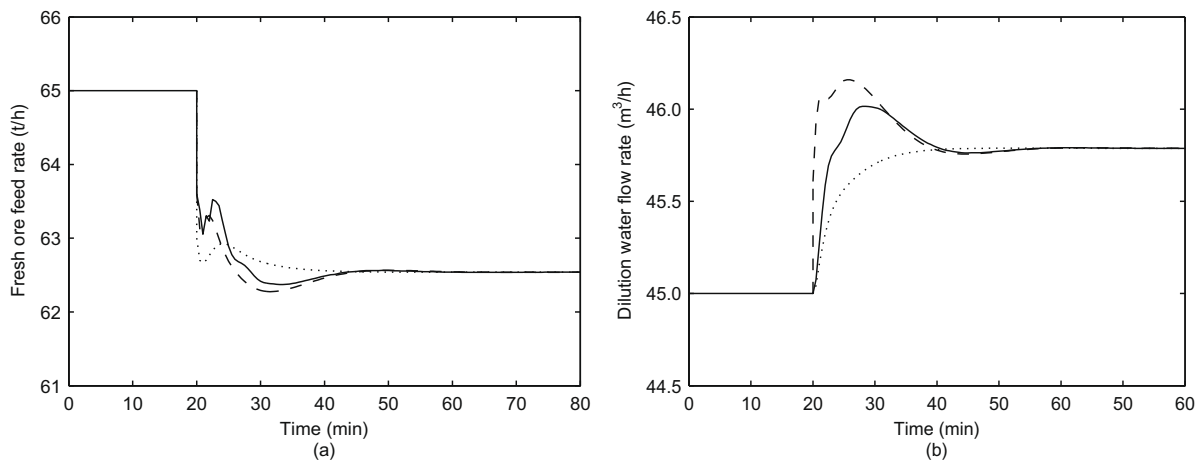


Fig. 7. Corresponding changes of manipulated variables for set-point change in particle size under DOMC (solid line), MPC (dash line) and PI (dot line) schemes in the presence of model mismatches: (a) fresh ore feed rate; (b) dilution water flow rate. The set-point change in particle size is from 70% to 71% – 200 mesh.

Comparing (24) with (25), we can find that severe model mismatches exist here. Set-point tracking for product particle size and circulating load are simulated in the case of model mismatches respectively in this part.

Case I: Set-point change of the particle size in the case of model mismatches.

In *Case I*, the response curves of the controlled variables for set-point change in particle size under DOMC, MPC as well as PI schemes are shown in Fig. 6. The set-point of the product particle size is from 70% to 71% – 200 mesh at $t = 20$ min. The corresponding changes of the manipulated variables, namely the fresh ore feed rate and the dilution water flow rate, are shown in Fig. 7. It can be seen from Fig. 6a that the settling time of the particle size

under the proposed DOMC method is about 16 min, which is much shorter than that under the MPC and multi-loop PI schemes (about 30 min). Fig. 6b shows that the three methods almost have the same settling times, but the proposed method has the fastest convergence speed and the smallest amplitude compared with the other two control schemes.

Performance indexes, including overshoot and integral of absolute error (IAE) are employed here to estimate the control performance of set-points tracking in the presence of model mismatches. Values of the performance indexes in *Case I* are listed in Table 2. It can be observed that the overshoots for both the particle size and the circulating load under the proposed scheme are smaller than that under the MPC and PI schemes. The IAE of the circulating load under the proposed scheme are much smaller than that under the other two control schemes.

Case II: Set-point change of the circulating load in the case of model mismatches.

Similarly, the response curves of the controlled variables for set-point change in circulating load are shown in Fig. 8. The set-point of the circulating load is from 150 to 155 t/h at $t = 20$ min. The corresponding changes of the manipulated variables are shown in Fig. 9. Fig. 8a shows that the DOMC scheme has a smaller

Table 2

Performance indexes in set-point change of particle size.

	Particle size y_1			Circulating load y_2		
	DOMC	MPC	PI	DOMC	MPC	PI
Overshoot (%)	0	0.04	0	0.26	0.29	0.54
IAE	4.21	3.68	4.36	3.17	5.80	9.85

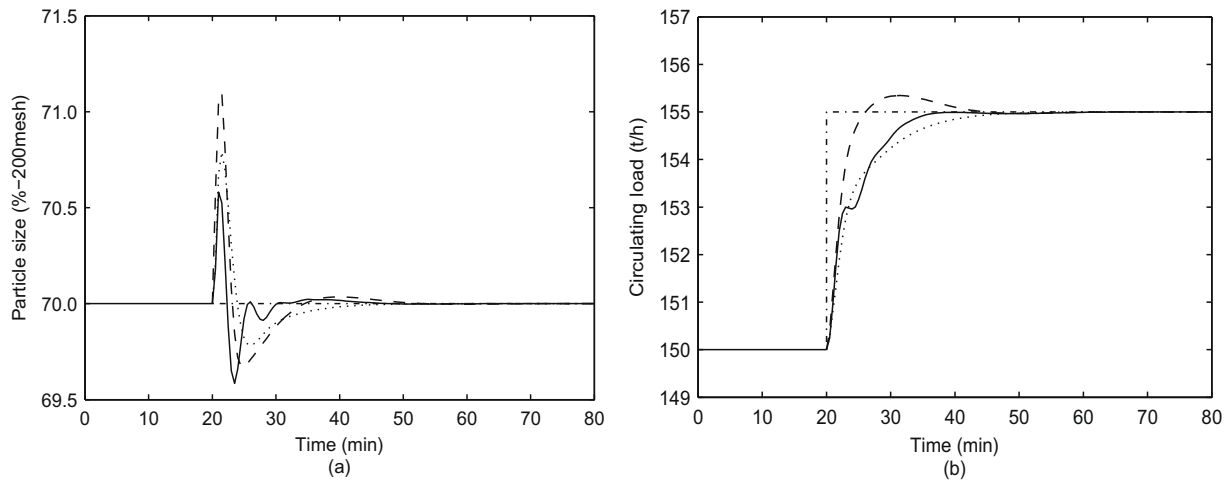


Fig. 8. Response curves of controlled variables for set-point change in circulating load under DOMC (solid line), MPC (dash line) and PI (dot line) schemes in the presence of model mismatches: (a) particle size; (b) circulating load. The set-point change (dash-dot line) in circulating load is from 150 to 155 t/h.

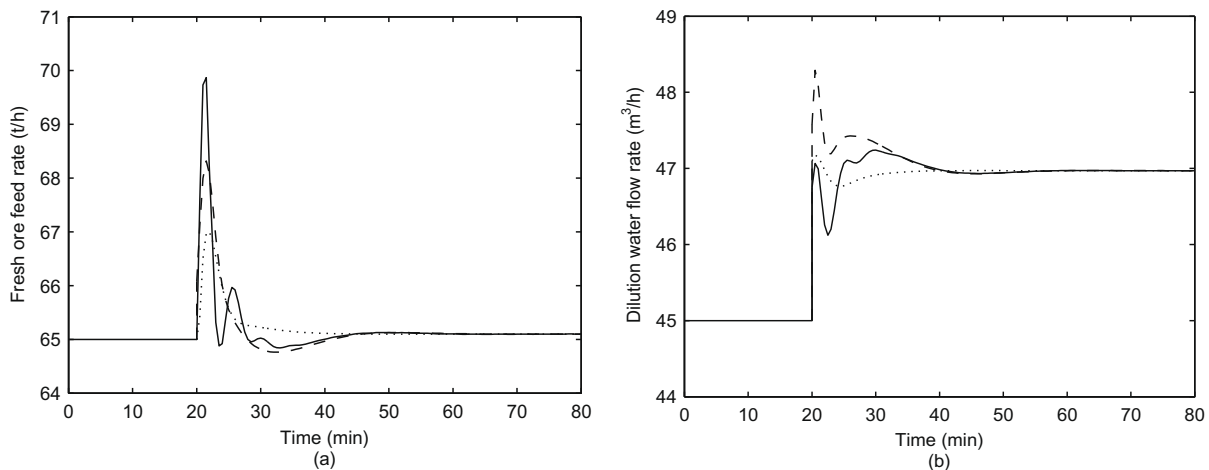


Fig. 9. Corresponding changes of manipulated variables for set-point change in circulating load under DOMC (solid line), MPC (dash line) and PI (dot line) schemes in the presence of model mismatches: (a) fresh ore feed rate; (b) dilution water flow rate. The set-point change in circulating load is from 150 to 155 t/h.

Table 3

Performance indexes in set-point change of circulating load.

	Particle size y_1			Circulating load y_2		
	DOMC	MPC	PI	DOMC	MPC	PI
Overshoot (%)	0.83	1.56	1.13	0	0.22	0
IAE	1.93	4.25	3.16	22.46	15.54	25.85

magnitude than that under the MPC and PI schemes. It can be observed from Fig. 8b that the settling time for response of the circulating load under the proposed method (about 16 min) is shorter than that under the MPC and PI schemes (about 25 min).

Performance indexes under the three control schemes in Case II are shown in Table 3. The overshoot for the product particle size under the DOMC scheme are smaller than that under the MPC scheme. The overshoots for both the product particle size and the circulating load under the proposed scheme are the smallest compared with that under the MPC or PI schemes. IAE of the particle size under the proposed method is smaller than that under the other two methods.

4.2. External disturbance rejection

One of the most difficult problem for control system design in grinding circuit lies in the strong external disturbances, such as variations of ore hardness and feed ore particle size. These external disturbances bring about many problems when designing controller and tuning parameters, even resulting in unstable control of the process. According to [25], ore hardness has great influence on product particle size and circulating load. For example, ore feed with increased ore hardness leads to a decreased particle size (fraction of – 200 mesh) and an increased circulating load, and these dynamics can be modelled as a first-order plus dead time (FOPDT) form which is the most commonly used model to describe the dynamics of industrial process. For simulation study, assume the transfer functions of the process with the external disturbance are described as

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix} \begin{bmatrix} U_1(s) \\ U_2(s) \end{bmatrix} + \begin{bmatrix} H_1(s) \\ H_2(s) \end{bmatrix} D_{ex}(s), \quad (26)$$

$$D_{ex1}(s) = H_1(s)D_{ex}(s) = \frac{-0.32e^{-0.52s}}{4.8s + 1}D_{ex}(s),$$

$$D_{ex2}(s) = H_2(s)D_{ex}(s) = \frac{1.6e^{-0.46s}}{5.3s + 1}D_{ex}(s),$$

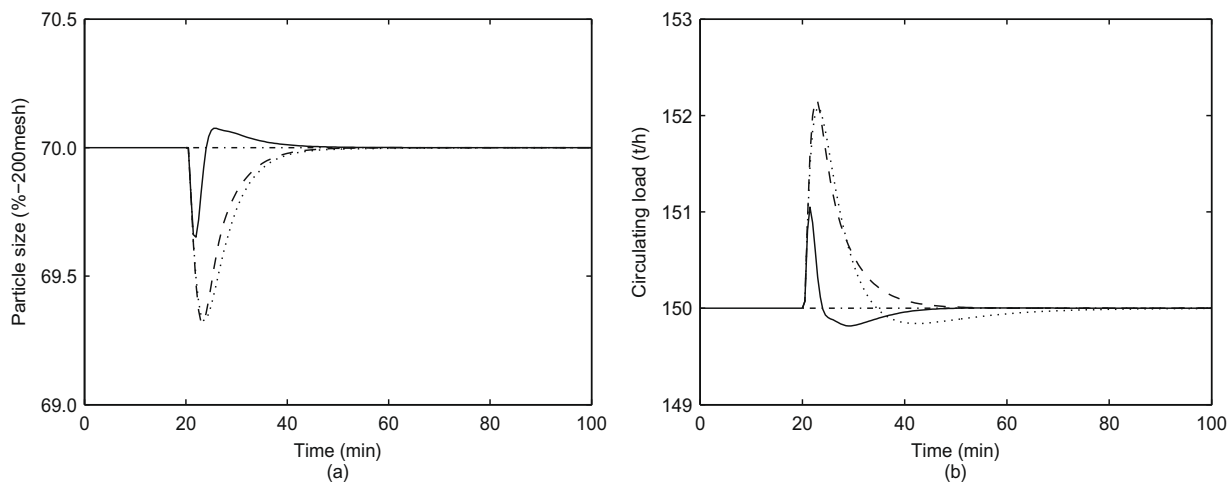


Fig. 10. Response curves of controlled variables in the presence of strong external disturbance under DOMC (solid line), MPC (dash line) and PI (dot line) schemes: (a) particle size; (b) circulating load. The set-points are denoted by dash-dot lines.

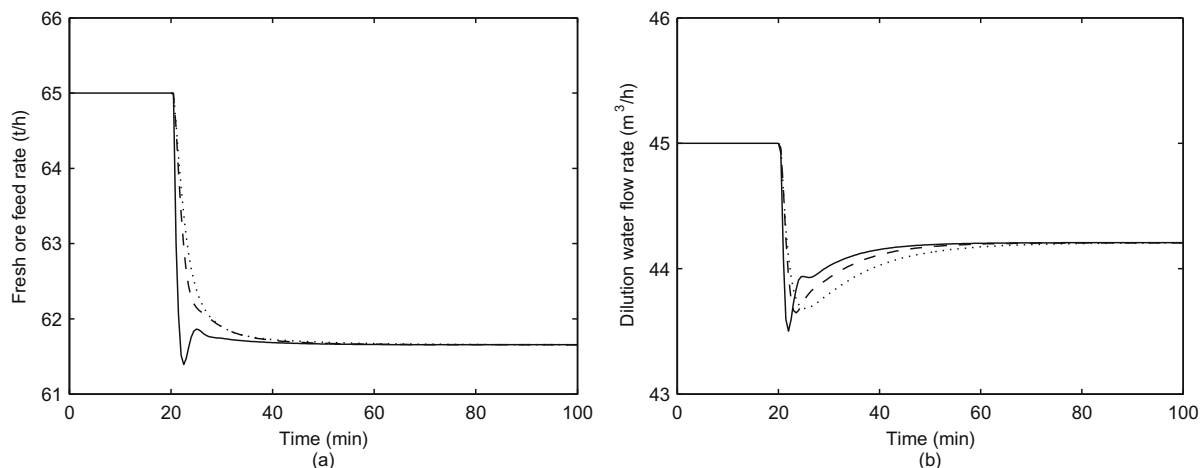


Fig. 11. Corresponding changes of manipulated variables in the presence of strong external disturbance under DOMC (solid line), MPC (dash line) and PI (dot line) schemes: (a) fresh ore feed rate; and (b) dilution water flow rate.

Table 4

Performance indexes in the presence of strong external disturbances.

	Particle size y_1			Circulating load y_2		
	DOMC	MPC	PI	DOMC	MPC	PI
Overshoot (%)	0.50	0.97	0.96	0.70	1.44	1.39
IAE	1.42	4.71	5.57	4.02	14.50	17.00

where $D_{ex}(s)$ represents the external disturbance (variation of ore hardness).

In this part, disturbance rejection ability of the proposed method is compared with both MPC and PI methods. Consider the case that a strong external disturbance is imposed on the grinding circuit: the ore hardness has an increase of +10% at $t = 20$ min. The closed-loop response curves of the product particle size and the circulating load are shown in Fig. 10. The corresponding changes of the manipulated variables are displayed in Fig. 11.

It can be seen from Fig. 10 that the dynamic behaviors of both the product particle size and the circulating load under the proposed method are much better than that under the MPC or PI scheme. Compared with MPC and PI schemes, the proposed method has the shortest settling time for both the particle size and the circulating load. Moreover, the magnitudes of both the particle size and the circulating load under the proposed method are much smaller than that under the MPC and PI schemes.

The performance indexes of strong external disturbance rejection under DOMC, MPC and PI schemes are shown in Table 4. It can be observed that the overshoots of both the product particle size and the circulating load under DOMC scheme are much smaller than that under MPC or PI scheme. The IAEs of both the particle size and circulating load under the proposed scheme are also much smaller than that under the MPC and PI schemes.

Here, to have better performances for the closed-loop systems under MPC scheme, we have selected large dimensions for P and M . However, this may cause a large computation burden. Even in this case, in the presence of model mismatches, our DOMC method shows a comparative performance with lower computation burden compared with that of MPC method. Moreover, in the presence of external disturbances, our method shows a better disturbance rejection property than those of the MPC and PI methods in controlling ball mill grinding circuits.

5. Conclusion

In grinding process, the disturbances include the external disturbances (such as variations of ore hardness and feed particle size) and internal disturbances caused by model mismatches as well as coupling effects. These disturbances will degrade the performance of ball mill grinding circuits if we do not consider a correspondingly designed controller for them. In this paper, a disturbance observer based multi-variable control (DOMC) strategy has been developed to control the ball mill grinding circuits (MIMO systems) with such external and internal disturbances. The control strategy consists of two compound controllers for the two loops, respectively. Each controller includes two parts: a PID feedback part plus a feed-forward compensation part based on disturbance estimation. A rigorous analysis has also been given to show the reason why the DOB can effectively suppress the disturbances. The simulation results have shown that compared with MPC and PI schemes, the closed loop system of ball mill grinding circuit under DOMC strategy possesses a better disturbance rejection property, no matter for the model mismatches or for the external disturbances.

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References

- [1] A. Pomerleau, D. Hodouin, A. Desbiens, E. Gagnon, A survey of grinding circuit control method: from decentralized PID controllers to multivariable predictive controllers, *Powder Technology* 108 (2) (2000) 103–115.
- [2] M. Ramasamy, S.S. Narayanan, C.D.P. Rao, Control of ball mill grinding circuit using model predictive control scheme, *Journal of Process Control* 15 (2005) 273–283.
- [3] O. Galan, G.W. Barton, J.A. Romagnoli, Robust control of a SAG mill, *Powder Technology* 124 (2002) 264–271.
- [4] K. Najim, D. Hodouin, A. Desbiens, Adaptive control: state of the art and application to a grinding process, *Powder Technology* 82 (1) (1995) 59–68.
- [5] A.V.E. Conradie, C. Aldrich, Neurocontrol of a ball mill grinding circuit using evolutionary reinforcement learning, *Minerals Engineering* 14 (2001) 1277–1294.
- [6] X.S. Chen, J.Y. Zhai, S.H. Li, Q. Li, Application of model predictive control in ball mill grinding circuit, *Minerals Engineering* 20 (2007) 1099–1108.
- [7] X.S. Chen, S.H. Li, J.Y. Zhai, Q. Li, Expert system based adaptive dynamic matrix control for ball mill grinding circuit, *Expert Systems with Applications* 36 (2009) 716–723.
- [8] X.S. Chen, Q. Li, S.M. Fei, Supervisory expert control for ball mill grinding circuits, *Expert Systems with Applications* 34 (2008) 1877–1885.
- [9] V.R. Radhakrishnan, Model based supervisory control of a ball mill grinding circuit, *Journal of Process Control* 9 (1999) 195–211.
- [10] X.S. Chen, Q. Li, S.M. Fei, Constrained model predictive control in ball mill grinding process, *Powder Technology* 186 (1) (2008) 31–39.
- [11] K. Ohishi, M. Nakao, K. Ohnishi, K. Miyachi, Microprocessor-controlled DC motor for load-insensitive position servo system, *IEEE Transactions on Industrial Electronics* 34 (1) (1987) 44–49.
- [12] W.H. Chen, P.J. Gawthrop, J.O. Reilly, A nonlinear disturbance observer for robotic manipulators, *IEEE Transactions on Industrial Electronics* 47 (4) (2000) 932–938.
- [13] S. Komada, N. Machii, T. Hori, Control of redundant manipulators considering order of disturbance observer, *IEEE Transactions on Industrial Electronics* 47 (2) (2000) 413–420.
- [14] S. Katsura, Y. Matsumoto, K. Ohnishi, Modeling of force sensing and validation of disturbance observer for force control, *IEEE Transactions on Industrial Electronics* 54 (1) (2007) 530–538.
- [15] L. Guo, M. Tomizuka, High-speed and high-precision motion control with an optimal hybrid feedforward controller, *IEEE Transaction on Mechatronics* 2 (2) (1997) 110–122.
- [16] J. Ishikawa, M. Tomizuka, A novel add-on compensator for cancellation of pivot nonlinearities in hard disk drives, *IEEE Transaction on Magnetics* 34 (4) (1998) 1895–1897.
- [17] Y.H. Huang, W. Messner, A novel disturbance observer design for magnetic hard drive servo system with rotary actuator, *IEEE Transaction on Magnetics* 34 (4) (1998) 1892–1894.
- [18] C.J. Kemf, S. Kobayashi, Disturbance observer and feedforward design for a high-speed direct-drive positioning table, *IEEE Transactions on Control Systems Technology* 7 (5) (1999) 513–526.
- [19] W.H. Chen, Nonlinear disturbance observer enhanced dynamic inversion control of missiles, *Journal of Guidance Control and Dynamic Systems* 26 (1) (2003) 161–166.
- [20] W.H. Chen, Disturbance observer based control for nonlinear systems, *IEEE Transactions on Mechatronics* 9 (4) (2004) 706–710.
- [21] M. Bertoluzzo, G.S. Buja, E. Stampacchia, Performance analysis of a high-bandwidth torque disturbance compensator, *IEEE Transactions on Mechatronics* 9 (4) (2004) 653–660.
- [22] L. Guo, W.H. Chen, Disturbance attenuation and rejection for a class of nonlinear systems via DOBC approach, *International Journal of Robust and Nonlinear Control* 15 (3) (2005) 109–125.
- [23] H. Kobayashi, S. Katsura, K. Ohnishi, An analysis of parameter variations of disturbance observer for motion control, *IEEE Transactions on Industrial Electronics* 54 (6) (2007) 3413–3421.
- [24] H.D. Zhu, G.H. Zhang, H.H. Shao, Control of the process with inverse response and dead-time based on disturbance observer, in: *American Control Conference*, Portland, OR, USA, June 8–10, 2005, pp. 4826–4831.
- [25] Q.H. Li, *Crushing and grinding*, Metallurgical Industry Press, Beijing, 1980.
- [26] W.R. Wang, S.X. Pan, F. Wang, A comparison study of two disturbance rejection control strategies for hydraulic position servo systems, in: *Proceedings of the 30th Annual Conference of the IEEE Industrial Electronics Society*, Busan, Korea, November 2–6, 2004, pp. 1993–1996.
- [27] T. Umeno, Y. Hori, Robust speed control of DC servomotors using modern two degrees-of-freedom controller design, *IEEE Transactions on Industrial Electronics* 38 (5) (1991) 363–368.